

Single-stage to orbit: The lunar case

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December 12, 2025

Abstract

This extended abstract summarizes the ascent modeling performed for the Apollo Lunar Module ascent stage using direct numerical integration in polar coordinates. A random program burn-time was generated as required, and the corresponding ascent trajectory, steering-law response, orbital-element survey (parametric sweep in ϕ_0), and fuel use were computed. Results reported below distinguish single-run quantities (for $\phi_0 = 40^\circ$), the randomized burn-time used for validation checks, and the orbit-element values obtained from the full ϕ_0 sweep.

1 Introduction

The ascent of the Apollo Lunar Module is modeled by solving the coupled radial–angular equations of motion under thrust and lunar gravity. A linear–tangent steering program governs the pitch $\phi_{th}(t)$ evolution. Two kinds of computations were performed: (i) a single ascent simulation with $\phi_0 = 40^\circ$ used for time histories and validation checks, and (ii) a parametric sweep in ϕ_0 (range 1° – 70°) used to compute orbital elements at insertion and to select the initial angle that minimizes the squared error to the desired preliminary orbit.

2 Results Summary

A random burn-time was drawn in the notebook (seed-based): the random generator returned 448.095 s; for the final integration runs the total thrusting time used was

$$T_{\text{burn}} = 458.095 \text{ s},$$

with a vertical ascent phase of $t_{\text{vert}} = 10 \text{ s}$ in the vertical-to-linear–tangent steering law implementation. Single-run verification used $\phi_0 = 40^\circ$; the ϕ_0 sweep produced the orbital-element survey and the minimization result.

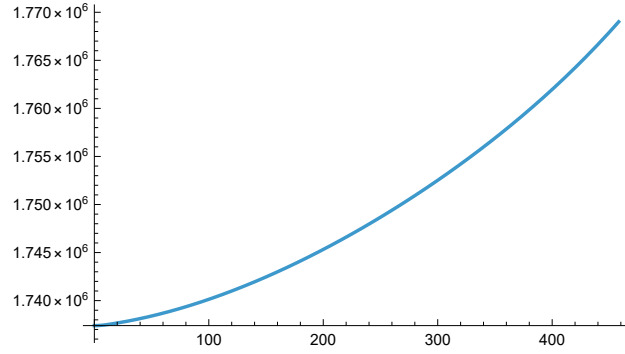


Figure 1: Radius versus time (single-run, $\phi_0 = 40^\circ$).

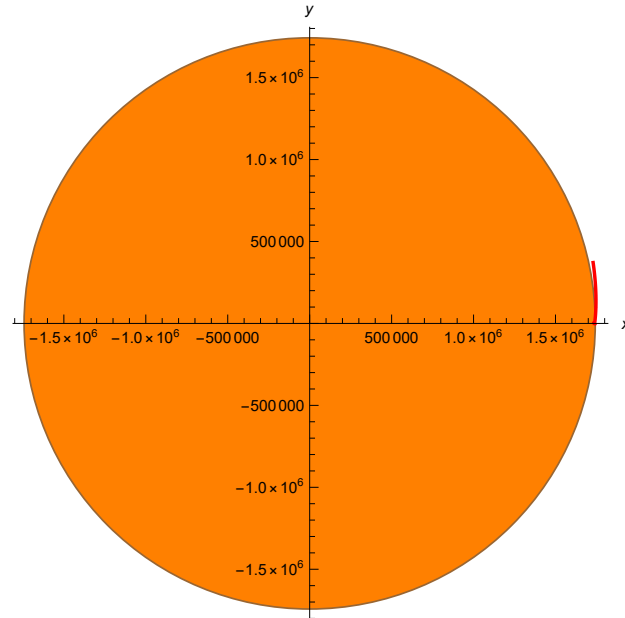


Figure 2: Ascent trajectory in Cartesian coordinates (single-run, $\phi_0 = 40^\circ$).

Validation checks (single-run, $\phi_0 = 40^\circ$): altitude at $t = 30$ s matches historical Apollo ascent records:

$$H(30 \text{ s}) \approx 484.385 \text{ m} \quad (1589.19 \text{ ft}).$$

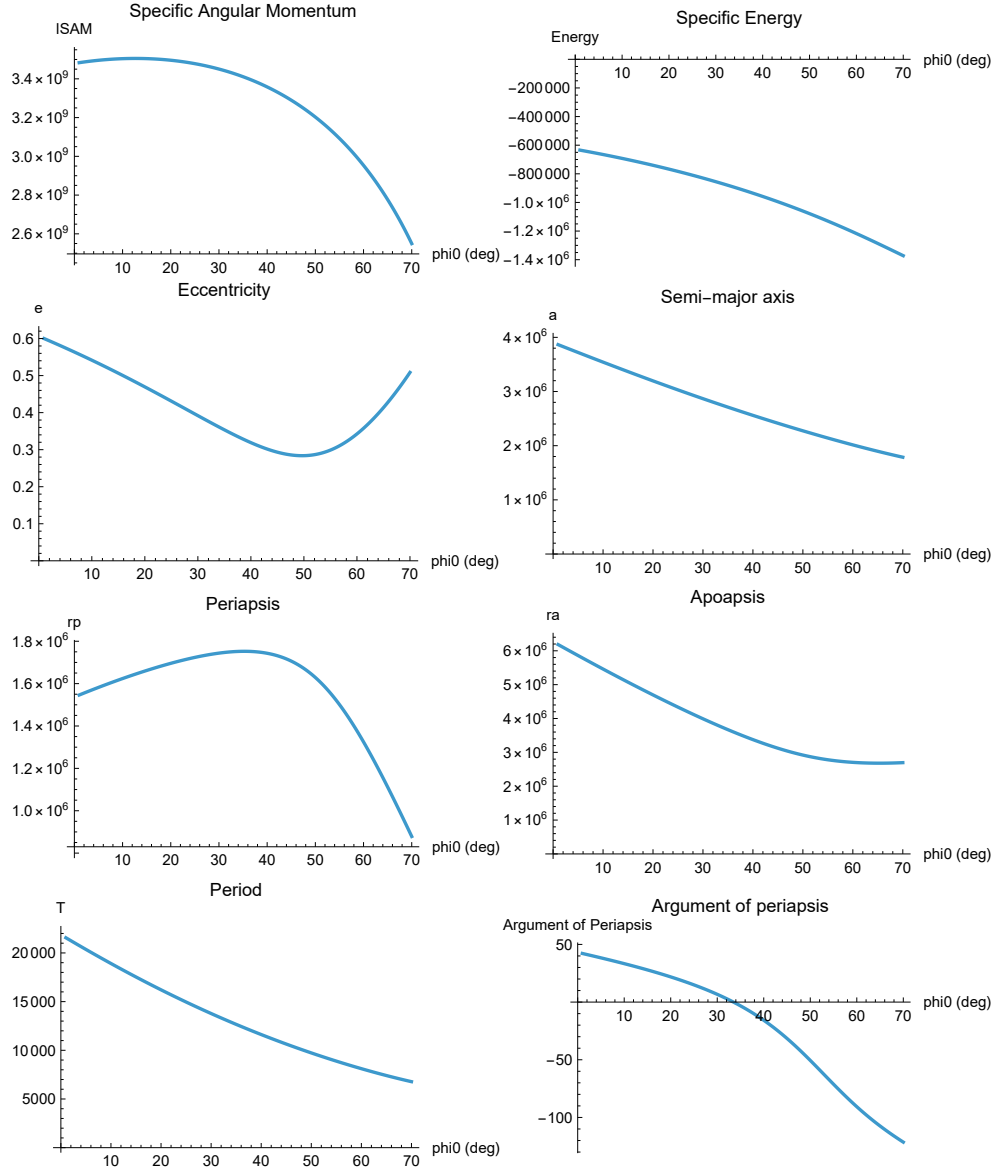


Figure 3: Collection of ascent orbital-element plots produced by the ϕ_0 sweep.

Sweep diagnostics: the ϕ_0 survey reported a *minimizing* initial angle

$$\text{best } \phi_0 = 58^\circ,$$

where the objective minimized was the squared deviation between computed periapsis/apoapsis and the desired preliminary values (periapsis altitude = 9 nmi, apoapsis altitude = 45 nmi). However, the sweep minimum ($\phi_0 = 58^\circ$) produces a periapsis radius *below* the lunar radius (i.e., $r_p < R_{\text{Moon}}$) and therefore is not an acceptable operational insertion. The sweep therefore must be interpreted with caution: the unconstrained error minimum does not automatically guarantee a physically safe insertion. For safe insertions (requiring $r_p > R_{\text{Moon}}$) the notebook finds the interval

$$28.2^\circ \leq \phi_0 \leq 41.1^\circ,$$

so the single-run choice $\phi_0 = 40^\circ$ lies within the safe interval and was used for time-history validation runs.

Reported insertion state at $t = T_{\text{burn}} = 458.095$ s:

$$r_{\text{ins}} = 1.76901 \times 10^6 \text{ m}, \quad \dot{r}_{\text{ins}} = 134.768 \text{ m/s}, \quad v_{t,\text{ins}} = 1928.2 \text{ m/s},$$

giving a resultant speed $|v| \approx 1932.9$ m/s and an altitude

$$H_{\text{ins}} \approx 31.6 \text{ km}.$$

Orbital elements computed from the ϕ_0 -survey The unconstrained minimum at $\phi_0 = 58^\circ$ yields:

$$\begin{aligned} r_p &= 1.40053 \times 10^6 \text{ m}, & r_a &= 2.72834 \times 10^6 \text{ m}, \\ e &\approx 0.1727, & a &\approx 2.11116 \times 10^6 \text{ m}, & T &\approx 8.70 \times 10^3 \text{ s}. \end{aligned}$$

Important interpretive note: the periapsis above is a *radius*, not an altitude; since the lunar radius is $R_{\text{Moon}} = 1.7374 \times 10^6$ m, the reported $r_p = 1.40053 \times 10^6$ m from the unconstrained minimize is *below* the surface (impact). This is why the safe-phi interval (above) is essential ; in operational planning the initial angle must be chosen inside the safe interval even if the unconstrained element error is smaller elsewhere.

Fuel consumption (single-run mass-flow estimate used for the time histories): with mass flow $\dot{m} \approx 5.10476$ kg/s (derived from thrust and I_{sp}) and the burn time used for the single-run,

$$m_{\text{fuel}} \approx \dot{m} \cdot T_{\text{burn}} \approx 5.10476 \text{ kg/s} \times 458.095 \text{ s} \approx 2.3385 \text{ tonnes.}$$

This numerical fuel consumption exceeds the ideal rocket-equation estimate (≈ 1.9975 t) because of gravity and steering losses and the non-optimal steering program used in many runs.

3 Operational Phases

Following Bennett, the ascent is divided into four operational phases; here we explicitly separate *validation-run* results (single-run $\phi_0 = 40^\circ$) from *sweep* diagnostics (selection of safe ϕ_0 interval).

(1) Vertical Liftoff (0–10 s). Thrust aligned with the radial direction ($\phi = 90^\circ$). Single-run vertical-phase results validate leap-off altitude at $t = 30$ s: $H(30 \text{ s}) \approx 484$ m, consistent with Apollo figures.

(2) Pitch-over and Tangent Steering. The linear tangent law pro-

gressively reduces ϕ from ϕ_0 to zero over the program time (time history plotted in the notebook). Single-run time histories at $\phi_0 = 40^\circ$ show the expected transfer of energy from radial to tangential components.

(3) Burnout and Orbit Insertion. Engine cutoff at $t = T_{\text{burn}}$. Orbital insertion element values are reported above; note that the unconstrained minimizer of the element-error yields $\phi_0 = 58^\circ$ but produces $r_p < R_{\text{Moon}}$ and therefore is not acceptable. Operational selection must therefore enforce $r_p > R_{\text{Moon}}$ (safe-phi interval: $28.2^\circ \leq \phi_0 \leq 41.1^\circ$).

(4) Free Coast. After shutdown the two-body dynamics is integrated to confirm periapsis and orbital period. For the single-run validation the orbital coast integration confirms the period $T \approx 8.70 \times 10^3$ s .

4 Extension to Chinese Robotic Missions

The numerical framework ported to Chang’e-class ascent stages requires replacing vehicle mass, dry mass, thrust and I_{sp} and adopting closed-loop guidance for ascent (rather than the fixed linear–tangent law). For sample-return missions targeting near-circular parking orbits (120–200 km), the sweep-and-minimize approach can be replaced with an optimal-control solver or closed-loop guidance to enforce rendezvous constraints and to ensure periapsis safety (i.e., $r_p > R_{\text{Moon}}$).

5 Conclusions

The notebook contains both single-run ascent histories for validation and a comprehensive ϕ_0 survey. The unconstrained error-minimizer of the orbital-element difference is $\phi_0 = 58^\circ$ (but it yields a periapsis below the lunar radius); enforcing a periapsis-safety constraint yields the safe interval $28.2^\circ \leq \phi_0 \leq 41.1^\circ$. For validation and figures we used $\phi_0 = 40^\circ$ and $T_{\text{burn}} = 458.095$ s; the single-run fuel use is $m_{\text{fuel}} \approx 2.3385$ t.

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